

Primjer kolokvija iz MATLAB-a

Zadatak 1:

- a) Izračunajte koristeći format rational $\left\{-\frac{8}{3}:\left(6-\frac{16}{3}\right)-\frac{4}{5}\left[12-\left(2-\frac{34}{29}\right)\cdot\frac{29}{12}\right]\right\}\cdot\left(-\frac{1}{6}\right)+\frac{1}{2}$
- b) Izračunajte koristeći format bank $\sqrt[4]{1.6\cdot 10^{-3}}+2\cdot\sqrt[3]{0.125}$
- c) Izračunajte :7!

Izračunajte : $\left(\sqrt{10}\right)^{-\log 25+2\log 55}$

Zadatak 2:

Za $A = \{1,2,3,4,5,6\}$, $B = \{x: x \in \mathbb{N}, x \text{ je djeljiv s } 2 \text{ i } x \leq 10\}$ i
 $C = \{x: x \in \mathbb{N}, x \text{ je prost broj i } x < 10\}$, elementi univerzalnog skupa
 $\mathbb{N} = \{1,2,3,4,5,6,7,8,9,10,11,12\}$, odredite :

$$A \cup B, A^C, A \cap B, A \cap B^C, A \cap B \cap C, (A \setminus C) \cup B, (A \cup B)^C \cap C$$

Zadatak 3:

- a) Zbrojite, oduzmite, pomnožite i podijelite sljedeće polinome :
 $(x^6 - 5x^4 + 2x^3 - 3x^2 + x + 8)$ i $(x^4 + x^3 - 2x^2 - 5x + 4)$
- b) Odredite nul točke polinoma $x^4 + x^3 - 2x^2 - 5x + 4$
- c) Odredite koeficijente polinoma čija su rješenja : 1.5,2,4

Zadatak 4:

- a) Riješite matricnu jednadžbu $(A+X)\cdot(B-2\cdot I) = A$, ako je I jedinična matrica drugog reda
te $A = \begin{bmatrix} -1 & 8 \\ 3 & 4 \end{bmatrix}$ i $B = \begin{bmatrix} 4 & -5 \\ 2 & 6 \end{bmatrix}$.
- b) Kreirajte matricu A kao retčanu s elementima 0,5,10,...,45, zatim jednostupčanu matricu B kao transponiranu matricu matrice A. Zatim kreirajte jednoretčanu matricu C kvadrata elemenata matrice A te matricu D čiji će stupci bit elementi matrice B i C.

Zadatak 5:

Riješite sustav:

$$\begin{array}{rrrrrr} -x_1 & + & 2x_2 & & + & 3x_4 & = & 7 \\ 2x_1 & & & + & 3x_3 & + & x_4 & = & -3 \\ x_1 & - & 2x_2 & - & x_3 & - & x_4 & = & 4 \\ x_1 & + & x_2 & & & + & 2x_4 & = & 7 \end{array}$$

Zadatak 6:

Grafički predstaviti funkciju: $f(x) = \sqrt{x^2 - 2x + 5}$

Zadatak 7:

Zadan je vektor $\overrightarrow{AD} = (1, -2, 3)$ Odredite koordinate točke D ako je $\vec{r}_A = -2\vec{i} + 3\vec{j} - 4\vec{k}$

Zadatak 8:

Izračunajte u skupu kompleksnih brojeva: $z = -10 \cdot \frac{(1-i)(2-3i)+i}{(2-i)(1-i)} - 7i$

Rješenja:

Zadatak 1:

format rational

```
>> (- (8/3) / (6 - 16/3) - (4/5) * (12 - (2 - 34/29) * (29/12))) * (-1/6) + 1/2
```

$$\left\{ -\frac{8}{3} : \left(6 - \frac{16}{3} \right) - \frac{4}{5} \left[12 - \left(2 - \frac{34}{29} \right) \cdot \frac{29}{12} \right] \right\} \cdot \left(-\frac{1}{6} \right) + \frac{1}{2}$$

ans =

5/2

format bank

```
>> (1.6*10^-3)^(1/4)+2*(0.125)^(1/3)
```

$$\sqrt[4]{1.6 \cdot 10^{-3}} + 2 \cdot \sqrt[3]{0.125}$$

ans =

1.20

```
>> format rational
```

```
>> factorial(7)
```

7!

ans =

5040

```
>> (sqrt(10))^(-log10(25)+2*log10(55))
```

$$\left(\sqrt{10} \right)^{-\log 25 + 2 \log 55}$$

ans =

11

Zadatak 2:

```
>> A=[1,2,3,4,5,6];B=[2,4,6,8,10];C=[2,3,5,7];N=[1,2,3,4,5,6,7,8,9,10,11,12];
```

```
>> union(A,B)
```

$A \cup B$

ans =

1 2 3 4 5 6 8 10

```
>> setdiff(N,A)  $A^c$ 
      7      8      9      10      11      12
```

```
>> intersect(A,B)  $A \cap B$ ,
```

```
ans =
```

```
      2      4      6
```

```
>> intersect(A,setdiff(N,B))  $A \cap B^c$ 
```

```
ans =
```

```
      1      3      5
```

```
>> intersect(A,intersect(B,C))  $A \cap B \cap C$ 
```

```
ans =
```

```
      2
```

```
>> union(setdiff(A,C),B)  $(A \setminus C) \cup B$ 
```

```
ans =
```

```
      1      2      4      6      8      10
```

```
>> intersect(setdiff(N,union(A,B)),C)  $(A \cup B)^c \cap C$ 
```

```
ans =
```

```
      7
```

Zadatak 3:

a)

```
>> p=[1,0,-5,2,-3,1,8];q=[1,1,-2,-5,4];q1=[0,0,1,1,-2,-5,4];
```

```
>> p+q1
```

```
ans =
```

```
      1      0      -4      3      -5      -4      12
```

```
>> p-q1
```

```
ans =
```

```
      1      0      -6      1      -1      6      4
```

```
>> conv(p,q)
```

```
ans =
```

```
1      1      -7      -8      13      19      -15      29      -33      -36      32
```

```
>> [d, R] = deconv(p,q)
```

```
d =
```

```
1      -1      -2
```

```
R =
```

```
0      0      0      7      -16      -5      16
```

b)

```
>> format short
```

```
>> f=[1,1,-2,-5,4];roots(f)
```

```
ans =
```

```
-1.5846 + 1.1487i
```

```
-1.5846 - 1.1487i
```

```
1.4482 + 0.0000i
```

```
0.7211 + 0.0000i
```

c)

```
>> format rational
```

```
>> r=[1.5,2,4];poly(r)
```

```
ans =
```

```
1      -15/2      17      -12
```

Dakle traženi polinom je: $x^3 - \frac{15}{2}x^2 + 17x - 12$.

Zadatak 4:

a)

$A = [-1, 8; 3, 4]; B = [4, -5; 2, 6];$

$\gg X = A * \text{inv}(B - 2 * \text{eye}(2)) - A$

$X =$

$-1/9 \quad -133/18$

$-25/9 \quad -49/18$

b)

$A = 0:5:45$

$A =$

0 5 10 15 20 25 30 35 40 45

$\gg B = A'$

$B =$

0

5

10

15

20

25

30

35

40

45

```
>> C=A.^2
```

```
C =
```

```
    0    25    100    225    400    625    900    1225    1600
2025
```

```
>> D=[B C']
```

```
D =
```

```
    0     0
    5    25
   10   100
   15   225
   20   400
   25   625
   30   900
   35  1225
   40  1600
   45  2025
```

Zadatak 5:

```
A=[-1,2,0,3;2,0,3,1;1,-2,-1,-1;1,1,0,2];b=[7;-3;4;7];
```

```
>> x=A\b
```

```
x =
```

```
    1
   -2
   -3
    4
```

```
>> rref([A,b])
```

```
ans =
```

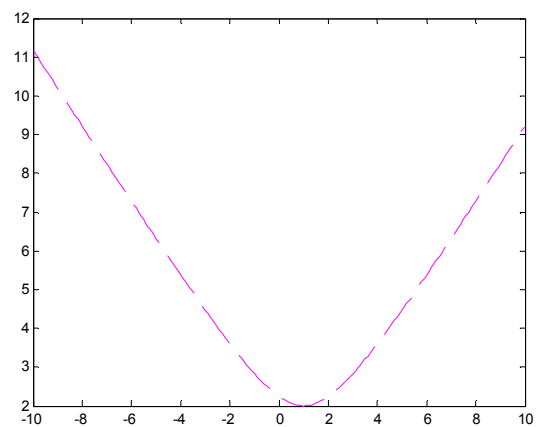
1	0	0	0	1
0	1	0	0	-2
0	0	1	0	-3
0	0	0	1	4

Zadatak 6:

```
x=-10:0.1:10;
```

```
>> y=sqrt(x.^2-2*x+5);
```

```
>> plot(x,y,'m--')
```



Zadatak 7:

```
>> ad=[1,-2,3];a=[-2,3,-4];
```

```
>> d=ad+a
```

d =

-1	1	-1
----	---	----

Zadatak 8:

```
>> format short
```

```
>> z=-10*((1-i)*(2-3*i)+i)/((2-i)*(1-i))-7*i
```

z =

-11.0000 - 0.0000i